

Introduction

Borrelia burgdorferi (*B. burgdorferi*) is a spirochete bacterial species. Their genome consists of a 900 kb linear chromosome along with several circular and linear plasmids. It is known that viruses called bacteriophages infect bacteria and can horizontally transfer genes between bacterial species. Spirochetes that cause Lyme disease like *B. burgdorferi* are known to carry cp32 prophage plasmids. Along with containing bacteriophage proteins, they also contain outer membrane lipoprotein genes like ospE which aid in immune evasion. When induced in the tick midgut via bloodmeal, these prophages are transcriptionally active and produce the components to form the virion ϕ BB-1 or phiBB-1. Bacteriophages are known to package and transfer mainly phage genomes like the cp32 prophage across hosts but research has indicated they may transfer bacterial genomes as well. Prior to this publication, the method of horizontal transfer for *B. burgdorferi* was unclear but hypothesized to involve general transduction through phiBB-1. The authors performed several experiments and analyzed the packaged DNA in phiBB-1 capsid heads via bioinformatic methods. They were able to identify that along with packaging the cp32 plasmids, phiBB-1 virions also package components of the *B. burgdorferi* genome such as the chromosome and lp54 plasmids (1).

Figure 5 analyzes the raw sequence reads of the DNA found in phiBB-1 bacteriophage capsids after being induced *in vitro* in a *B. burgdorferi* CA-11.2A culture. Additionally, the raw data was published for public access by the authors. The aim of this reconstruction project is to replicate Figures 5B, 5C, and 5D which are created using this data and following the Methods section. Some aspects such as Figure 5A are outside of the scope of this paper and require additional tools. This project serves as a culmination and demonstration of what has been learned while participating in an Independent Study in a lab that uses bioinformatic pipelines to analyze Lyme genomics (2).

Methods

Original Acquisition of Experimental Data

Figures 5B, 5C, and 5D were generated using raw sequencing data acquired from bacteriophages that packed *B. burgdorferi* DNA. To induce phiBB-1, *Borrelia burgdorferi* B31 were grown in BSK-II growth medium that allowed phage production. Based on the known background, it was expected for the phages to package the host genome. After allowing 72 hours for phage production, the culture was centrifuged to isolate a supernatant that contained the bacteriophages while bacterial cells were removed. The phage supernatants were purified and precipitated for DNA extraction. DNAase and high molecular weight salmon sperm DNA was added to remove any background DNA present that was not directly packaged by the phage. Phage capsids were digested with proteinase K and the DNA present in the capsids was purified using an ethanol method. This was repeated for a total of 3 replicates. The purified phage DNA was sequenced using a MinION MK1-B along with sequencing libraries and a barcoding kit. Further workup included using Guppy for basecalling and adaptor trimming before the raw read data was submitted to NCBI.

Published Computational Pipeline

According to the “Sequence analysis pipeline” of the Methods section of the publication, Filtlong was used to isolate reads over 5 kb and a quality score over 7. Read length refers to the number of bases read in a particular DNA fragment that was sequenced. Longer reads are more reliable across repetitive regions which is key for accurate genome alignment when plasmids are present (3). A quality score determines how likely each base call in a read is incorrect. A higher quality score is directly proportional to a more accurate base call. Based on the phred quality score method, a score of 10 is 90% accurate or a 1 in 10 probability of a base call being incorrect. A Q20 is 99% accurate, and a Q30 is 99.9% accurate (4) After the first round of filtering, each replicate was aligned to either a de novo assembled genome or the *B. burgdorferi*

CA-11.2A reference genome (RefSeq assembly: GCF_000172315.2) using the tool minimap2. Further filtering after mapping to the reference genome includes primary alignments with a MAPQ score over 20. Primary alignments refers to the best score alignment of a read after genome alignment, filtering out lower score secondary and supplemental reads. MAPQ score also uses the phred quality scoring method but is used to assess confidence of read placement during genome alignment (5). Lastly coverage maps were made of the primary mapping or the primary and supplemental mapping reads aligned to either the de novo assembled genome or the reference genome. Final files were created using Samtools and Seqkit and viewed using R and ggplot2.

Reconstruction Pipeline

The reconstructed workflow was designed to recreate Figures 5B, 5C, and 5D as accurately as possible with publicly available data. The de novo assembled genome was excluded due to the complexities of assembling a de novo genome using raw data. To reconstruct this pipeline, the reference genome was located in NCBI using the Refseq assembly number. Raw sequencing data was acquired in the listed SRA links. The long reads of the bacteriophage sequencing was done using the Oxford Nanopore MinION (6). The reference genome was downloaded as a FASTA file while the replicates were downloaded as FASTQ files (7-10). The raw reads were filtered as described using Filtlong (v0.3.1)(11). There was a total of three replicates. Each replicate was individually aligned to the CA-11.2A reference genome using minimap2 (2.30-r1287)(12) and a SAM file was created. Samtools (1.23.1)(13) was used for further workup including converting to a BAM file, sorting the file, filtering by mapping and MAPQ, indexing, and creating coverage tables. Additionally, raw read data was extracted from the filtered and sorted BAM file and converted to final files for assessing read length. Each of the final files had accession numbers referencing the components of the packaged genome but wasn't explicitly labeled. Another final format including the labeled components was generated from the reference genome FASTA file. This workflow was performed using a conda environment (25.11.1) on a macOS Sonoma 14.7.7 Terminal. The commands can be found under

the bash script “seqpipe.sh”. Although SeqKit was used in the original methods, it was excluded here due to no clear direction as to which filtering functions were done.

Further workup was done on R (4.5.2, RStudio 2026.1.0.392). The packages used include tidyverse (2.0.0)(14), ggplot2 (4.0.2)(15), ggbreak (0.1.7)(16) and knitr (1.51)(17). Final file formats were aligned to genome labels and organized to include all three replicates to accurately view the datasets involved in Figures 5B, 5C, and 5D. The commands to generate the figures can be found under the Rmd file “BB1Fig5BCD”.

Results

Reconstruction of Figure 5B

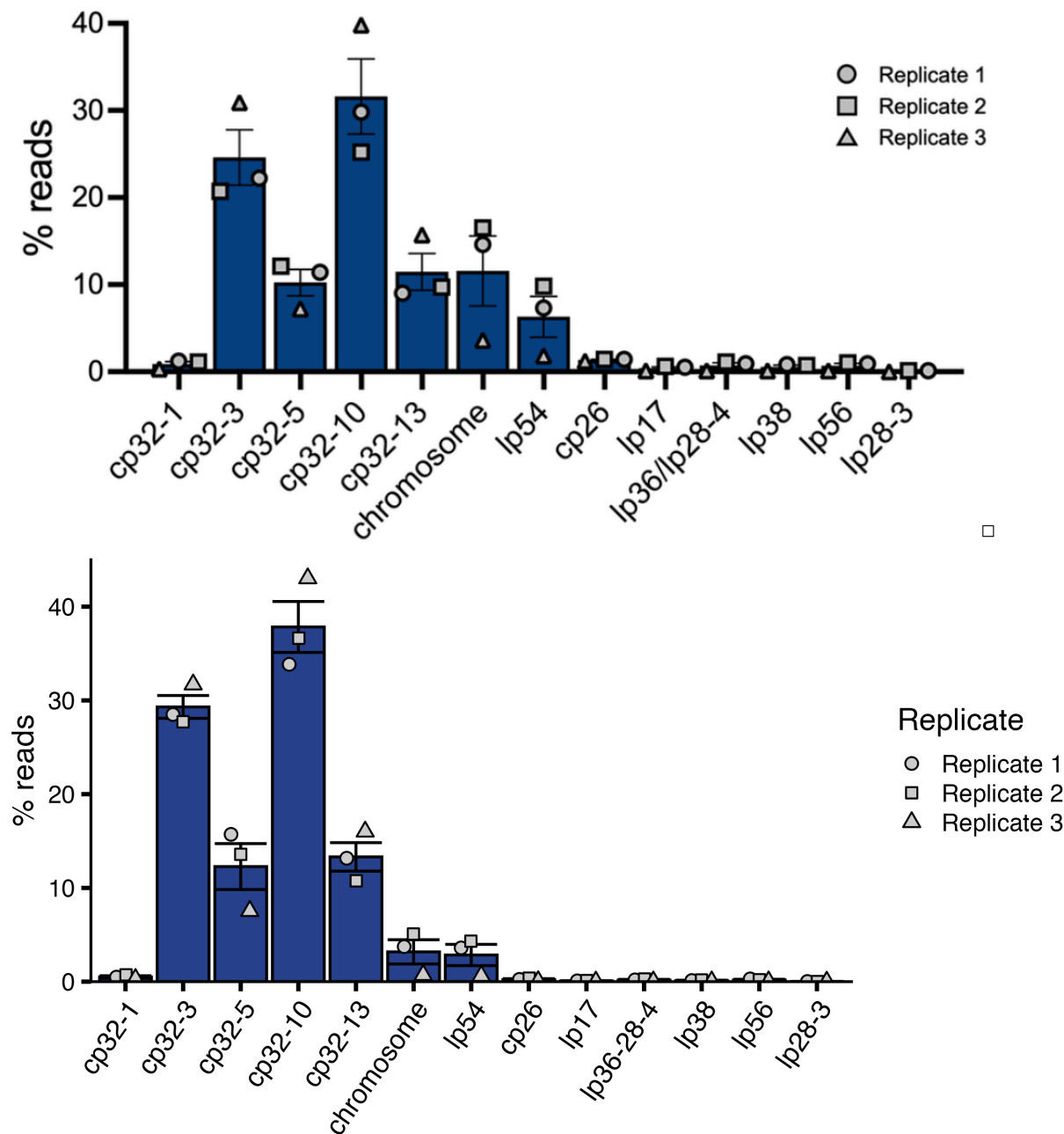


FIGURE 1:
TOP: ORIGINAL FIGURE 5B
BOTTOM: RECONSTRUCTED

The three figures discussed derives from the DNA that was sequenced after phiBB1 phages were produced and incubated in a culture of *Borrelia burgdorferi*. They were then isolated, purified of background DNA, and lysed to extract DNA stored in the capsid. The three figures provide strong evidence that the phiBB-1 bacteriophage does acquire and package the *B. burgdorferi* genome. Figure 5B quantifies what percent of the read data of each replicate corresponds to a specific plasmid isoform or the *B. burgdorferi* chromosome. All three replicates share similar results as well. Upon analysis, it can be concluded that the phiBB1 virions preferentially package various isoforms of the cp32 plasmids along with packaging smaller portions of other parts of the genome like the chromosome and lp54 plasmid.

Directly comparing the original figure to the reconstructed figure, the general trend of preferential packaging of cp32 is conserved but the results have dissimilarities. In the original figure, Cp32-10 covers 32% and cp32-3 covers 25% of all reads. Cp32-13, cp32-5, and the chromosome are around 10% each while lp54 is around 7.5%. The cp32 isoforms cover about 79% of all reads. In the reconstructed figure, the cp32 isoforms covered about 92% of all reads. Cp32-10 covers 34%, cp32-3 covers 28%, cp32-13 covers 13% and cp32-5 covers 15%. The chromosome and lp54 covers around 4% each.

In comparison to the original, the reconstruction produced higher packaging percent reads for each of the cp32 isoforms while having lower percent reads for the chromosome and lp54. Likely reasons to account for the discrepancy may be due to not using Seqkit for filtering or knowing the exact parameters used. Another possibility is if the original figures are based on aligning to the *de novo* assembled genome as opposed to the reference genome. Although the publication states long reads were used to map to a genome, they stated that mapping was done to both the reference and the *de novo* assembled genome and does not specify which mapping was used in context to the figures.

Reconstruction of Figure 5C

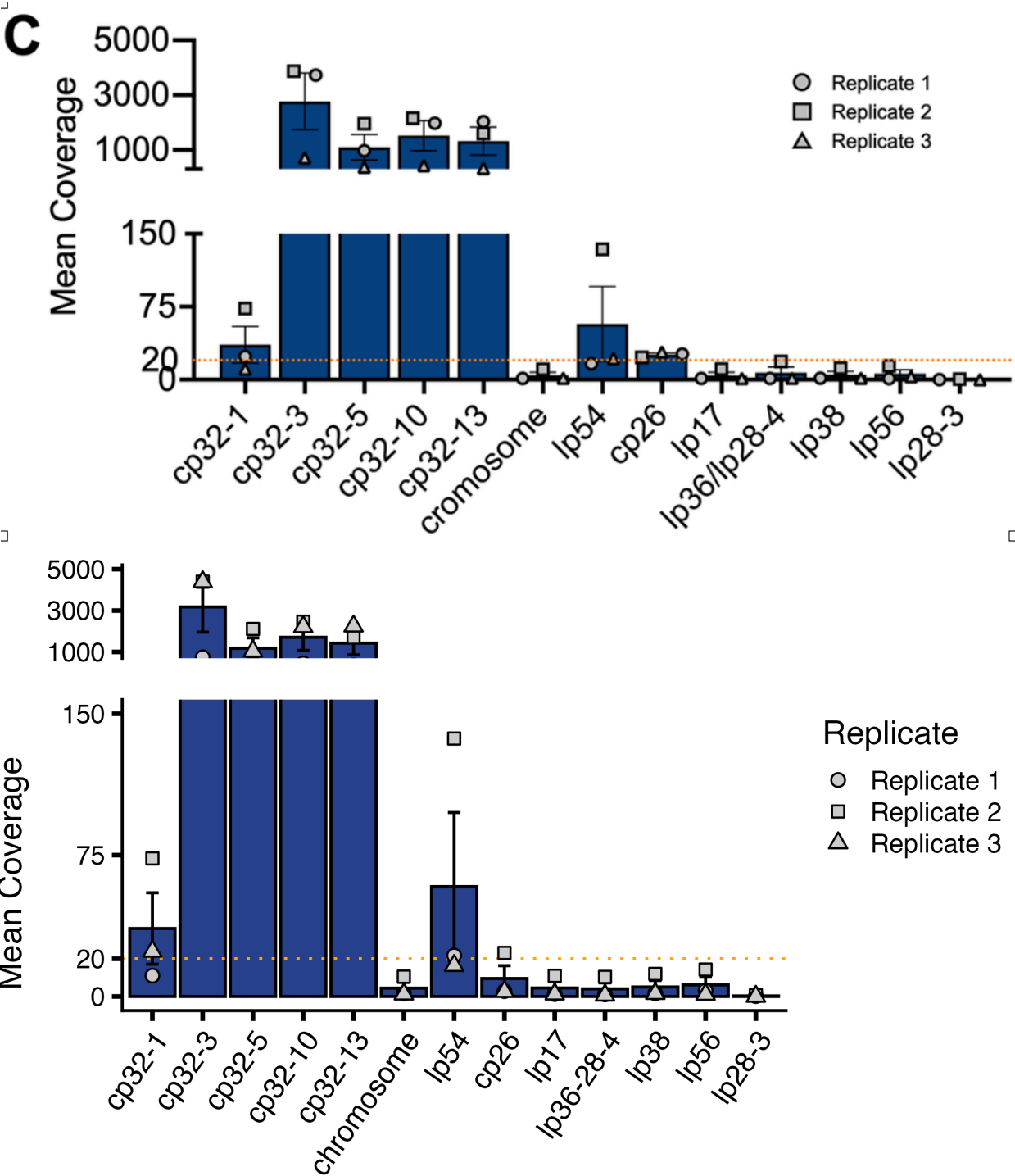


FIGURE 2:
TOP: ORIGINAL FIGURE 5C
BOTTOM: RECONSTRUCTED

Figure 5C uses the same coverage tables as Figure 5B to assess mean coverage. For this figure, mean coverage was calculated using the column referred to as mean depth. It can be understood as the number of times a fragment that was sequenced covered a region of DNA that corresponds to one of the plasmids or chromosome of the *Borrelia burgdorferi* genome. It is expected that the regions with the most coverage would be similar to the regions that have the highest percent reads as seen in Figure 5B. If a region of DNA is overwhelmingly present in contrast to other regions, then the amount of times coverage overlap occurs should be proportional. Upon observation, the cp32 plasmid isoforms have the highest mean coverage, spanning from 1000x to 5000x coverage while the other components range from 20-50x in comparison, correlating to Figure 5B but the trends as to which cp32 isoform dominates does not replicate the results of Figure 5B.

The order for highest to lowest coverage was anticipated to be cp32-10, cp32-3, cp32-5, and cp32-13 based on a correlation to percent read order. The chromosome and lp54 regions were expected to have lower coverage. However, the original figure shows that cp32-3 has the highest coverage, followed by cp32-10, cp32-13, and cp32-5. In addition the cp32-3 replicates ranged from 1000x to 4000x coverage while the other isoforms consistently ranged from 1000-1500x coverage, skewing the highest mean coverage. Although this finding is unexpected, it still supports the idea that phiBB1 preferentially package cp32 isoforms as opposed to other components of the genome.

The reconstructed Figure 5C preserved the major enrichment patterns observed in the original figure. Unlike Figure 5B, there are no strong differences in the coverage depths between the two. The difference in the y-axis of each figure is likely due to using the ggbreak package to recreate the broken y-axis while the package used in the original is not noted or may have been ggplot2.

However, a trend that is notable in both figures is the placement of the replicates. For example, the original Figure 5B has replicate 3 as the highest values, followed by replicate 1, and

then replicate 2. The reconstruction has the same observation for replicate 3 but is then followed by replicate 2 and replicate 1. Interestingly, a different pattern is seen in Figure 5C. The original has similar values for replicates 1 and 2 or has replicate 2 higher than the others. Replicate 3 tends to be at the bottom of the observed mean coverage across replicates. However, in the reconstruction of Figure 5C, replicate 2 conserves its trends while it seems as if replicates 1 and 3 have their positions swapped. Both figures have a trend between the original and reconstruction as one replicate being conserved while the other two are swapped but are different across the two figures. One possible explanation that can be drawn from this discrepancy is that coverage tables used to construct Figure 5B is different than Figure 5C in the original publication. In contrast, the reconstructed figures used the same coverage table. If the trends were consistent between the original and reconstruction, it would have been more likely a human error in deciding which long read dataset corresponds to its respective replicate. Despite the replicates being depicted differently across the original and the reconstruction, the preservation of the overall trends provides strong evidence towards the validity of the datasets used for reconstruction.

Reconstruction of Figure 5D

D

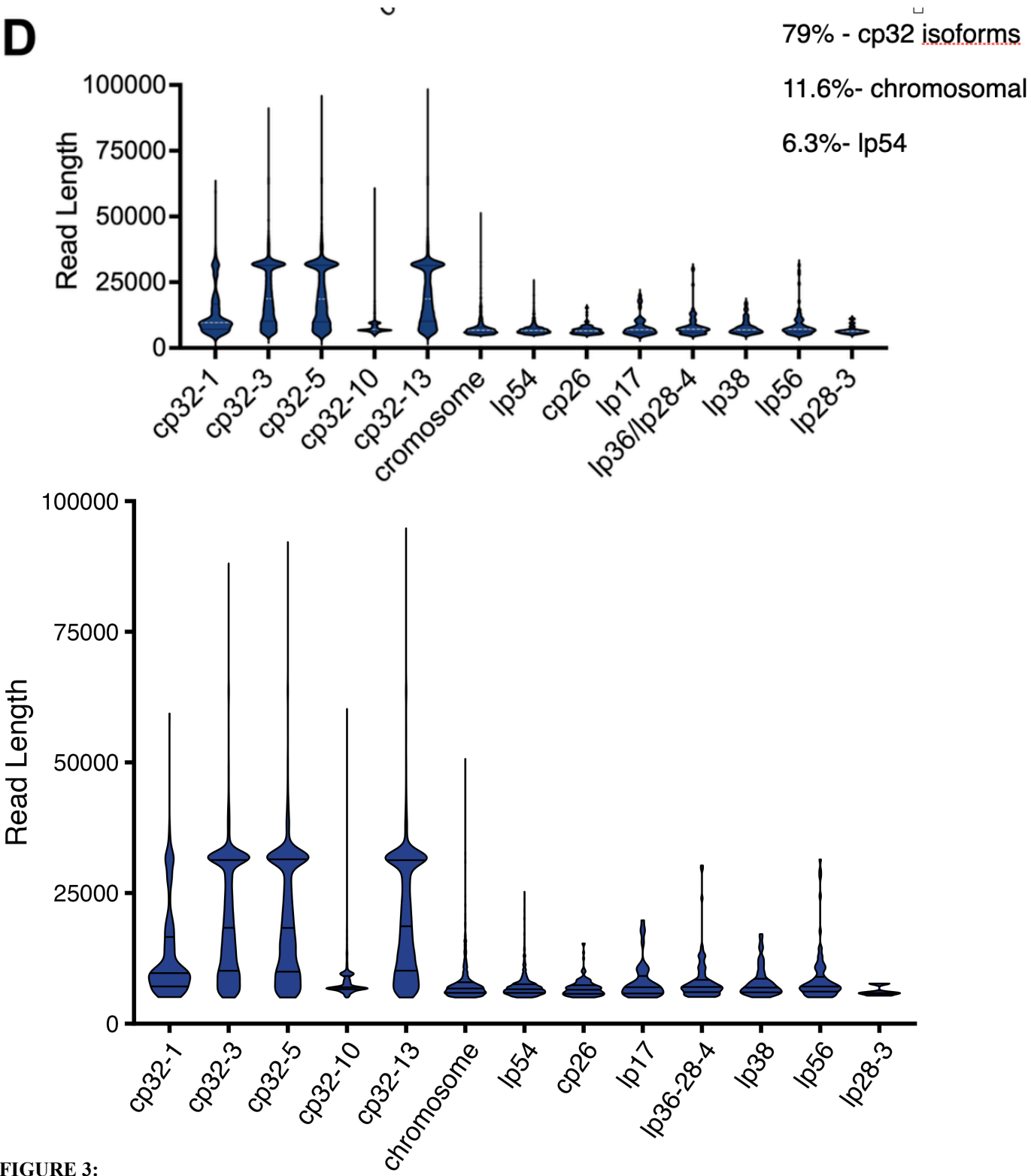


FIGURE 3:
TOP: ORIGINAL FIGURE 5D
BOTTOM: RECONSTRUCTED

While Figures 5B and 5C used the same coverage table, Figure 5D required a different dataset and pools the replicate data into one coverage table. The prior figures answer the questions as to which genomic component of *Borrelia burgdorferi* are packaged by phiBB-1 virions and how often they are packaged. Figure 5D focuses on how long each of the fragments that were being sequenced are. Instead of summarized coverage values, each of the fragments corresponding to a specific genomic component was accounted by read length to generate a violin plot. Between all three replicates, there were a total of 58,000 reads of various lengths identified and mapped.

The y-axis corresponds to the length of the acquired reads and the horizontal component in each of the columns correspond to how frequently read lengths were identified to be of a certain size. The cp32 isoforms had the longest read lengths and had strong density around the 30-35k read length in terms of the cp32-3, cp32-5 and cp32-13 isoforms. The cp32-1, cp32-10 and other plasmid components along with the chromosome often had shorter read lengths, around 8000 bp according to the publication.

While cp32-1 also correlates to prior figures due to having a low percent read and a low mean coverage along with short read lengths, cp32-10 directly contradicts prior data and changes observed trends. In Figure 5B, the cp32-10 isoform was identified to be the plasmid that was most frequently present in packaged DNA of phiBB-1 virions. While it didn't have the highest mean coverage, it was still among the highest and similar to other cp32 isoforms. Based on the prior trends, it was expected for cp32-10 to also have similar read lengths to the other frequently packaged and high coverage isoforms. However, it had much shorter read lengths despite the highest percent reads. Up to this point, the observable trend indicated that phiBB-1 bacteriophages preferentially package cp32 plasmid isoforms with certain isoforms being preferred over others. Frequently represented isoforms tend to also have longer read lengths when packaged while less preferred genomic components have smaller read lengths. The discrepancy of cp32-10 was not extensively discussed in the publication and could warrant further investigation as to why a frequently packaged cp32 isoform is packed in read lengths similar to a less preferred isoform like cp32-1.

The authors imply that percent reads, mean coverage, and read lengths may reflect the transcriptional activity of *B. burgdorferi*. This was supported by the observations that cp32-1 is not undergoing lytic replication due to its low packaging frequency. Additionally, a cp32-9 isoform in a B31 strain had similar observations as the cp32-1 isoform in the CA-11.2A strain and was known to be transcriptionally inactive compared to other cp32 isoforms. Further studies could extrapolate this trend to other non-cp32 components. However, it's likely the chromosome and lp54 plasmid are transcriptionally active due to higher packaging frequency and mean coverage in comparison to cp32-1 despite sharing similar read length. Based on this information, the cp32-10 plasmid read lengths are an outlier in the observed trends and read lengths may not be enough to fully explain packaging preferences.

In terms of the reconstructed Figure 5D, it is visually similar to the original figure and preserved the same read lengths across the genomic components. It does not provide any new information or contradictions to the original figure, supporting that the same datasets were used along with the same ggplot commands to generate the figure.

Conclusion

The results of this figure analysis lead to the general idea that phiBB-1 virions preferentially package cp-32 prophage associated genomes and some isoforms are more likely to be found than others. However, it is also evident that they package non-cp32 components that correspond to the *B. burgdorferi* genome which is a significant contribution to our current understanding of general transduction and mechanisms of horizontal gene transfer in this species. The bioinformatic methods used and the reconstruction pipeline successfully recreated the three figures discussed. Figure 5B had minor differences while the other two had minimal to no differences in data and only differed in visualization of the data. The probable rationale behind the differences in the cp32 isoforms along with *B. burgdorferi* genomic components identified by percent read is a difference in filtering method in generating coverage tables. As theorized before, Figure 5B, 5C, and 5D likely used the same datasets but had different coverage tables

down to the replicates. However despite the minor differences noted, the significant biological trends that support the central idea of this publication is preserved.

If this project was to be continued, the next steps would be to recreate Figure 5A which involves using BLAST and Kraken for taxonomic identification of bacterial species. Additionally, the *de novo* genome could be assembled using Trycycler, Flye, and Raven as described in the Methods. The raw reads would be aligned to this genome and the reconstruction pipeline would be repeated to compare the figures based on which genome the packaged DNA was aligned to. If the experiments done were to be extrapolated to novel research, another possibility would be to recreate the experimental settings in-vivo. *Ixodes scapularis* that contain *B. burgdorferi* CA-11.2A could be induced to produce phiBB-1 virions that are then extracted to analyze its packaged contents. This would give a comparison between an *in silico*, *in vitro*, and an *in vivo* model. Additionally, the contradictions found in the cp32-10 isoform could also be investigated to identify as to why its raw read lengths are similar to isoforms that are not lytically active but they are preferentially packaged by phiBB-1.

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